Relations

A function $f\colon A \to B$ maps elements from set A to elements in another set B.

We will now study more general *relationships* between elements than simple mappings.

Examples:

- "parenthood" amongst the set of people
  ("x is parent of y")
- "divisibility" amongst the set of integers
  ("x divides evenly into y")
- relationships between different sets:
  "enrolment" relating the sets of students and courses
  ("student s takes course c")
- relationships between more than two sets:
  "grade" relating students, courses and grades
  ("students s got a grade of g in course c")

Relations from Predicates

Example: Consider the predicate

$$\mathrm{Mark}(x, y, z) = \text{“student } x\text{, in course } y\text{, scored grade } z\text{”}$$

Let S, C and G be the sets of students, courses, and grades, respectively. The truth set of Mark is a subset Grades of the Cartesian product $S \times C \times G$ given as

$$\mathrm{Grades} = \{\, (s, c, g) : \mathrm{Mark}(s, c, g) \,\}$$

An n***-ary relation*** R is a subset of n-tuples.

If $R \subseteq A_1 \times A_2 \times \ldots \times A_n$, we say that

R *is a relation over* $A_1 \times A_2 \times \ldots \times A_n$.

E.g., Grades is a ternary (that is, 3-ary) relation over $S \times C \times G$.

IMDb Database – James Bond Films

	Title	Year	Starring Role	Screenwriters	Director
r01	Dr. No	1962	Sean Connery	Ian Fleming, Richard Maibaum	Terence Young
r02	From Russia with Love	1963	Sean Connery	Richard Maibaum	Terence Young
r03	Goldfinger	1964	Sean Connery	Richard Maibaum, Paul Dehn	Guy Hamilton
r04	Thunderball	1965	Sean Connery	Kevin McClory, Jack Whittingham	Terence Young
r05	You Only Live Twice	1967	Sean Connery	Roald Dahl, Harold Jack Bloom	Lewis Gilbert
r06	Casino Royale	1967	David Niven	Wolf Mankowitz, John Law, Michael Sayers	Val Guest
r07	On Her Majesty's Secret Service	1969	George Lazenby	Simon Raven, Richard Maibaum	Peter R. Hunt
r08	Diamonds Are Forever	1971	Sean Connery	Richard Maibaum, Tom Mankiewicz	Guy Hamilton
r09	Live and Let Die	1973	Roger Moore	Tom Mankiewicz	Guy Hamilton
r10	The Man with the Golden Gun	1974	Roger Moore	Richard Maibaum, Tom Mankiewicz	Guy Hamilton
r11	The Spy Who Loved Me	1977	Roger Moore	Christopher Wood, Richard Maibaum	Lewis Gilbert
r12	Moonraker	1979	Roger Moore	Christopher Wood	Lewis Gilbert
r13	For Your Eyes Only	1981	Roger Moore	Richard Maibaum, Michael G. Wilson	John Glen
r14	Octopussy	1983	Roger Moore	George MacDonald Fraser, Richard Maibaum	John Glen
r15	Never Say Never Again	1983	Sean Connery	Kevin McClory, Jack Whittingham	Irvin Kershner
r16	A View to a Kill	1985	Roger Moore	Ian Fleming, Richard Maibaum	John Glen
r17	The Living Daylights	1987	Timothy Dalton	Richard Maibaum, Michael G. Wilson	John Glen
r18	Licence to Kill	1989	Timothy Dalton	Michael G. Wilson, Richard Maibaum	John Glen
r19	GoldenEye	1995	Pierce Brosnan	Jeffrey Caine, Bruce Feirstein	Martin Campbell
r20	Tomorrow Never Dies	1997	Pierce Brosnan	Bruce Feirstein	Roger Spottiswoode
r21	The World Is Not Enough	1999	Pierce Brosnan	Neal Purvis, Robert Wade	Michael Apted
r22	Die Another Day	2002	Pierce Brosnan	Neal Purvis, Robert Wade	Lee Tamahori
r23	Casino Royale	2006	Daniel Craig	Neal Purvis, Robert Wade	Martin Campbell
r24	Quantum of Solace	2008	Daniel Craig	Paul Haggis, Neal Purvis	Marc Forster
r25	Skyfall	2012	Daniel Craig	Neal Purvis, Robert Wade	Sam Mendes
r26	Spectre	2015	Daniel Craig	John Logan, Neal Purvis	Sam Mendes
r27	No Time to Die	2021	Daniel Craig	Neal Purvis, Robert Wade, Cary Joji Fukunaga	Cary Joji Fukunaga

The IMDb Database

A small part of the Internet Movie Database (IMDb) representing James Bond Films is presented on the previous slide.

The database can be viewed as a 5-ary relation:

$$\mathrm{IMDb007} \subseteq \mathrm{Titles} \times \mathbb{N} \times \mathrm{Names} \times \mathcal{P}_{\mathrm{fin}}(\mathrm{Names}) \times \mathrm{Names}.$$

where Titles = set of film titles in the database

Names = set of names of people in the database

An element in the database associates a title with a year of release, the actor playing James Bond, its screenwriters, and its director.

For example

$$\mathrm{r10} = \Big(\text{The Man With the Golden Gun},\ 1974,\ \text{Roger Moore},\ \{\text{Richard Maibaum, Tom Mankiewicz}\},\ \text{Guy Hamilton} \Big).$$

Binary Relations

Binary (that is, 2-ary) relations are most common, often written in *infix* notation, i.e., as $a\,R\,b$ instead of $(a,b) \in R$; for example, $x = y$ instead of $=(x,y)$.

Binary relations are common for expressing:

- order ("a comes before b")
- equivalence ("a is the same as b")
- functions ("a maps to b")

A relation $R \subseteq A \times B$ is said to be a relation *from* A *to* B;

the set A is the ***source*** of R, and the set B is its ***sink***.

If $R \subseteq A \times A$ (ie, the source and sink are the same) then R is a relation *on* A.

Examples:

- The *less-than-or-equal* relation $\leq \subseteq \mathbb{N} \times \mathbb{N}$:

$$\leq \;=\; \{\,(x,y) : x \leq y\,\} \;=\; \{(0,0),\,(0,1),\,(1,1),\,(0,2),\,(1,2),\,(2,2),\,\ldots\}$$

- A relation $R \subseteq B \times \mathbb{N}$ from the set B of all living British people to the set $\mathbb{N}$ of natural numbers: $R = \{\,(b,n) : n \text{ is a bank account number of } b\,\}$.

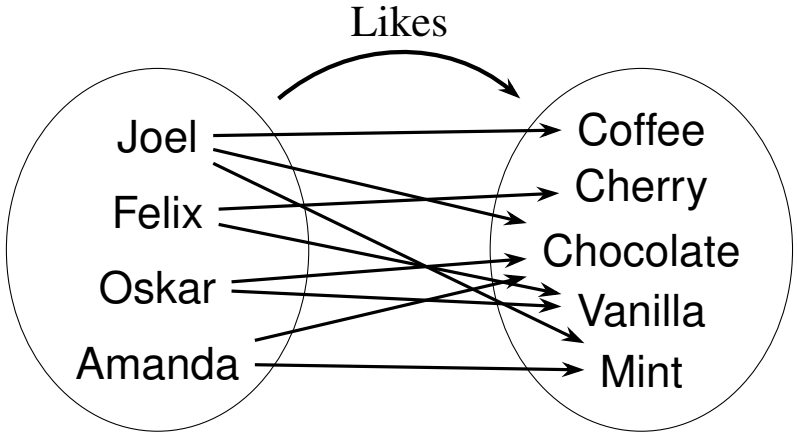
Binary Relations

Example: Joel likes mint, chocolate and coffee ice cream; Felix likes vanilla and cherry ice cream; Oskar likes vanilla and chocolate ice cream; and Amanda likes mint and chocolate ice cream. These properties define a binary relation

$$\begin{aligned} \text{Likes} \subseteq \text{Children} \times \text{Flavours} \qquad & \textit{where} \\ \text{Children} = \{\, & \text{Joel, Felix, Oskar, Amanda} \,\} \qquad \textit{and} \\ \text{Flavours} = \{\, & \text{Mint, Vanilla, Chocolate, Cherry, Coffee} \,\} \end{aligned}$$

$$\begin{aligned} \text{Likes} = \{\, & \text{(Joel, Mint), (Joel,Chocolate), (Joel,Coffee), (Felix, Vanilla), (Felix, Cherry),} \\ & \text{(Oskar, Vanilla), (Oskar, Chocolate), (Amanda, Mint), (Amanda, Chocolate)} \,\}. \end{aligned}$$

We can visualise such a relation as follows:



Visualising Relations

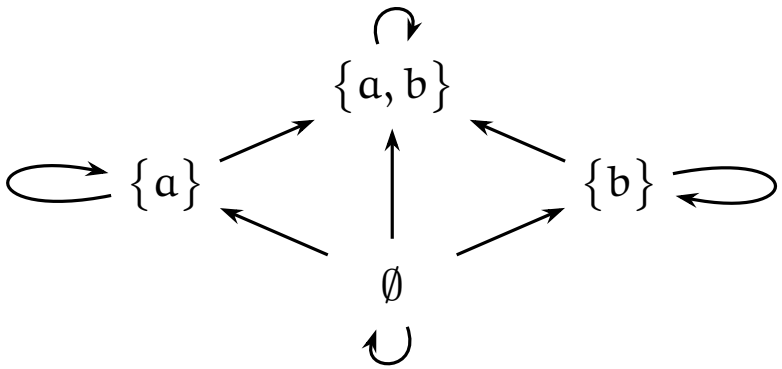
Binary relations from a set A to a set B can be visualised as above by drawing arrows from elements of the set A to related elements in the set B.

For homogeneous relations on a set A, we need only draw the set A once and add the relation arrows.

Example: The subset relation $\subseteq$ on the powerset of the set $\{a, b\}$:

$$\mathcal{P}\big(\{a, b\}\big) \;=\; \big\{\, \emptyset, \{a\}, \{b\}, \{a, b\} \,\big\}$$

is pictured as follows:



Operations on Binary Relations

A binary relation is a set $\mathrm{R} \subseteq \mathrm{A} \times \mathrm{B}$.

It thus makes sense to consider set operations on binary relations.

Example: Consider the binary relations Mother, Father and Parent on a set of people. For example, $\mathrm{Mother}(\mathrm{x}, \mathrm{y})$ indicates that x is mother of y. Then

- $\mathrm{Parent} = \mathrm{Father} \cup \mathrm{Mother}$
- $\mathrm{Father} = \mathrm{Parent} \setminus \mathrm{Mother}$

Examples: Consider the usual less than and equal to operations on numbers.

$$\leq \quad = \quad (<) \cup (=)$$

$$< \quad = \quad (\leq) \cap \overline{(=)}$$

$$= \quad = \quad (\leq) \setminus (<)$$

Inverse Relation

A binary relation $R \subseteq A \times B$ can be turned around to give a relation $R^{-1} \subseteq B \times A$ by considering the converse relation:

$$R^{-1} = \{ (b, a) : (a, b) \in R \}.$$

R^{-1} is referred to as the ***inverse*** of R

Examples:

- The inverse of the *less-than-or-equal-to* relation $\leq$ is the *greater-than-or-equal-to* relation $\geq$:

$$(\leq)^{-1} = \geq \quad \textit{since}$$

$$x \leq y \text{ if, and only if, } y \geq x$$

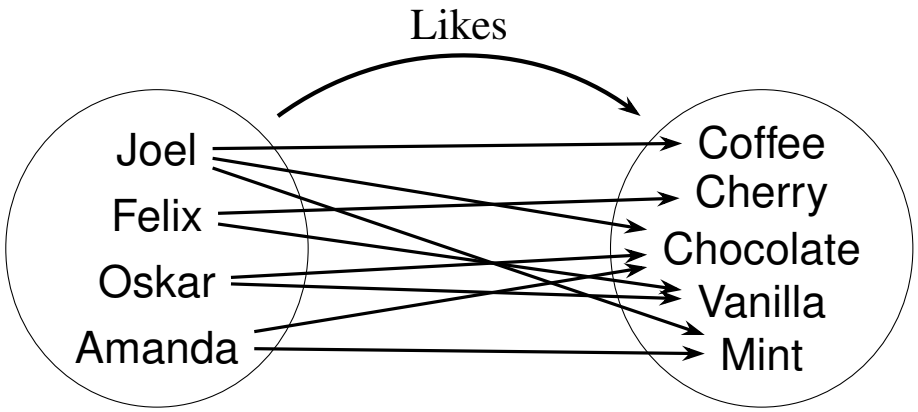
- The inverse of the Parent relation is the Child relation:

$$\mathrm{Parent}^{-1} = \mathrm{Child} \quad \textit{since}$$

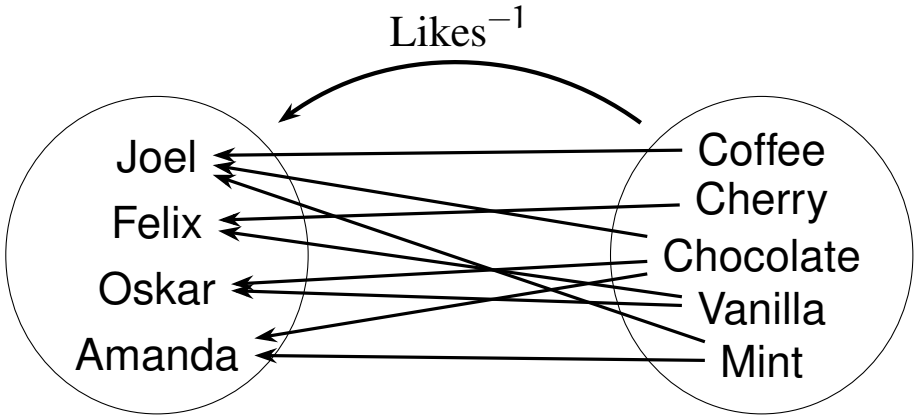
x is parent of y if, and only if, y is child of x

Inverse Relations

The inverse of the relation $\text{Likes} \subseteq \text{Children} \times \text{Flavours}$



is the relation $\text{Likes}^{-1} \subseteq \text{Flavours} \times \text{Children}$



For example, $(\text{Joel}, \text{Mint}) \in \text{Likes}$ indicates that Joel *likes* mint ice cream, whilst $(\text{Mint}, \text{Joel}) \in \text{Likes}^{-1}$ indicates that mint ice cream *is liked by* Joel.

Functions as Relations

The graph of a function $f : A \to B$ is a binary relation $R_f \subseteq A \times B$ in which every $a \in A$ is related to exactly one $b \in B$: $a \, R_f \, b$ if, and only if, $f(a) = b$.

Conversely, any binary relation $R \subseteq A \times B$ which satisfies this property is the graph of a function $f_R : A \to B$. $f_R(a) = b$ if, and only if, aRb.

Given a relation $R \subseteq A \times B$:

- the ***domain*** of R is the set
  $\text{domain}(R) = \{ a \in A : \exists b \in B \, (aRb) \}$
  `{a ∈ A: There exists b ∈ B s.t. (a R b)}`
- the ***range*** of R is the set
  $\text{range}(R) = \{ b \in B : \exists a \in A \, (aRb) \}$
  `{b ∈ B: There exists a ∈ A s.t. (a R b)}`

Note: $\text{domain}(R_f) = \text{domain}(f)$ and $\text{range}(R_f) = \text{range}(f)$.

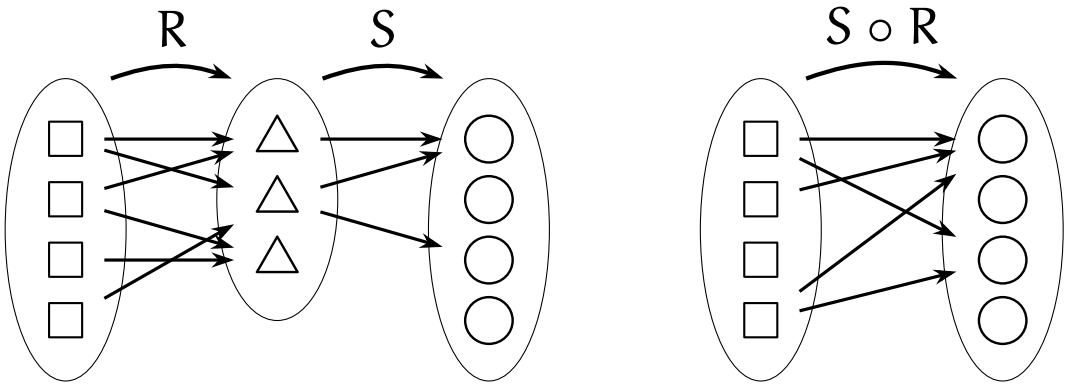
Example: Consider the relation Parent on humans:

- $\text{domain}(\text{Parent})$ = set of all parents (*not* all humans)
- $\text{range}(\text{Parent})$ = set of all children (i.e., *all* humans)

Composing Relations

Suppose $R \subseteq A \times B$ and $S \subseteq B \times C$ are binary relations in which the sink of R is the same as the source of S.

In such an instance, we can compose these relations one after the other, much like we compose functions.



The ***composition*** of S and R is the relation $S \circ R \subseteq A \times C$ gotten by composing one relation after the other:

$$S \circ R = \left\{ (a,c) \in A \times C : \exists\, b \in B \left((a,b) \in R \wedge (b,c) \in S \right) \right\}$$

{(a,c) ∈ A × C : There exists b ∈ B s.t. (a,b) ∈ R ∧ (b,c) ∈ S}

Example: A grandfather is a father of a parent: $\text{Grandfather} = \text{Parent} \circ \text{Father}$

Order matters: For example, a grandfather is a father of a parent, which is *not* the same as a parent of a father: $\text{Parent} \circ \text{Father} \neq \text{Father} \circ \text{Parent}$